

Week 16 - Audio-Only Deepfake Detection

Objective

Implement and evaluate an audio-only deepfake detection branch using MFCC features and a CNN architecture inspired by Khalid et al. (2021).

Methodology

- Dataset: FakeAVCeleb v1.2
- Audio preprocessing:
 - Audio extracted using FFmpeg
 - MFCC features computed (40 coefficients)
 - Saved as .npy tensors
- Model architecture:
 - ResNet-18 backbone
 - First convolution adapted for single-channel MFCC input
 - Binary output (real vs fake)
- Training setup:
 - Loss: BCEWithLogitsLoss
 - Optimizer: AdamW (lr = 1e-4)
 - Epochs: 5
 - CPU training

Results

- Accuracy: 97.20%
- F1-score: 98.55%
- AUC: 0.8785

Confusion Matrix:

```
[[ 42  58]
```

```
 [ 0 1975]]
```

Key observations:

Zero false negatives (no fake audio missed), strong bias towards detecting fake audio, and some real samples misclassified as fake.

Conclusion

The MFCC + ResNet18 audio branch demonstrates excellent performance on FakeAVCeleb, confirming that audio artefacts alone can strongly indicate deepfake manipulation. This validates the inclusion of an audio stream within the VerifAI multimodal framework and motivates future fusion with visual features.